

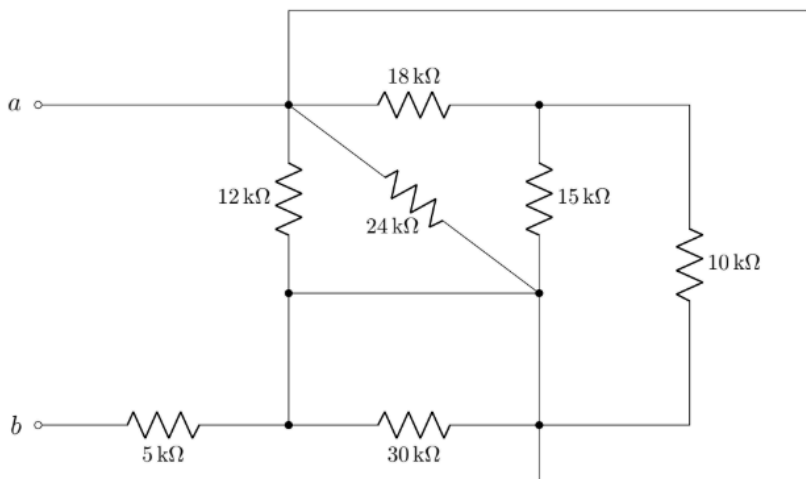
CSE250-10
Summer 26
ASSIGNMENT 1

Deadline: 06 July, 2026
(03:29 pm)

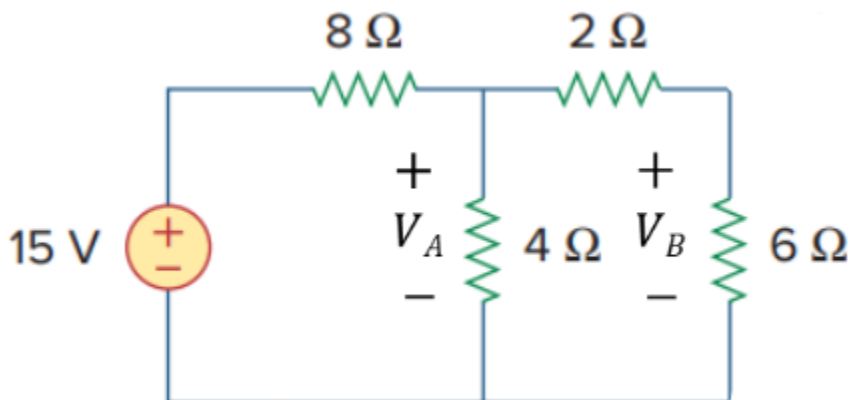
The goal of this assignment is to test your concept and knowledge of:

- Passive Sign Convention
- Current, Voltage and Power calculations
- Equivalent resistance
- VDR and CDR

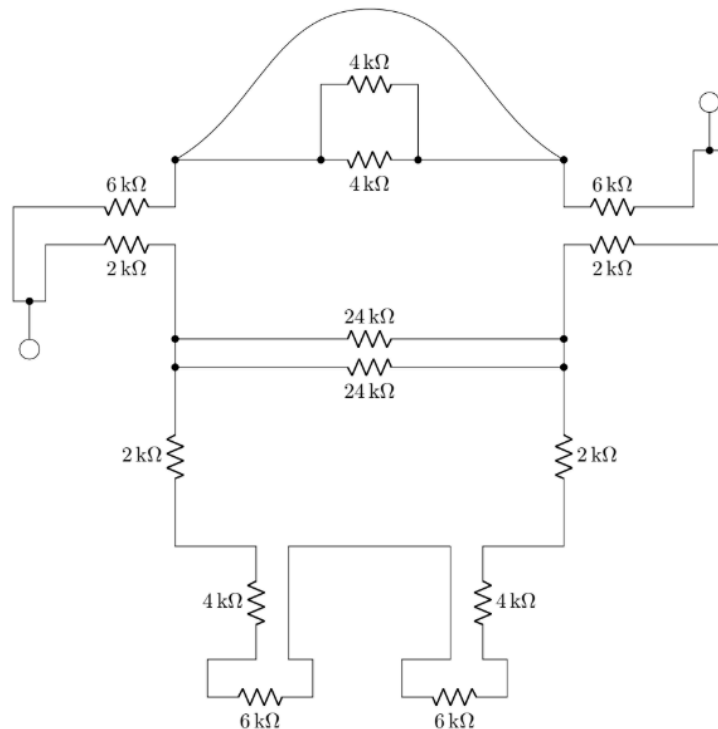
-
1. Determine R_{ab} , the equivalent resistance between the terminals a and b in the circuit shown below.



2. Find V_A , V_B using voltage division rule. Also find the current through 4 ohm and 2 ohm using current division rule.



3. Determine the equivalent resistance of the outermost nodes of the resistobot's hands, that is, between the circles in the given figure.



4. Apply KCL, KVL, and Ohm's Law to answer the following questions—
- [3 marks] Determine the current through the 9 V voltage source if the power of the element B1 is -12 W .
 - [3 marks] Determine I_x .
 - [2 marks] Determine V_x , the voltage across the 2 A current source.

